

Clinical Guideline: Asthma Anamnesis Support

Synthetic educational document for LangChain/Ollama RAG demonstration. This document is not a substitute for institutional protocols, national guidelines, or physician judgment.

1. Scope and Intended Use

This document provides a structured reference for generating an asthma-focused anamnesis draft. It is intended for software testing, prompt engineering, retrieval-augmented generation evaluation, and clinical documentation workflow prototyping. It should not be used as a standalone source for diagnosis, triage, or treatment.

2. Core Anamnesis Domains

Domain	Information to Collect
Chief complaint	Wheezing, cough, shortness of breath, chest tightness, nocturnal symptoms, exercise-induced symptoms.
History of present illness	Onset, duration, frequency, progression, episodic pattern, seasonal variation, and symptom-free intervals.
Triggers	Viral infections, allergens, exercise, cold air, smoke, occupational exposures, pets, dust, mold, medications, and emotional stress.
Severity indicators	Speech limitation, accessory muscle use, cyanosis, altered consciousness, inability to lie flat, reduced activity tolerance.
Past medical history	Previous asthma diagnosis, allergic rhinitis, eczema, recurrent bronchitis, pneumonia, hospital admission, ICU care, intubation.
Medication history	Short-acting bronchodilator use, inhaled corticosteroids, oral corticosteroids, leukotriene receptor antagonists, biologics, adherence, inhaler technique.
Allergies	Drug, food, environmental, latex, and contrast allergies. Include reaction type and severity where available.
Family history	Asthma, atopy, allergic rhinitis, eczema, chronic lung disease.
Social and environmental history	Smoking exposure, vaping exposure, indoor biomass exposure, occupational exposures, housing conditions, school or work absenteeism.
Review of systems	Fever, sputum, chest pain, reflux symptoms, nasal congestion, snoring, anxiety, weight loss, cardiac symptoms.

3. Red-Flag Features Requiring Urgent Clinical Review

- Severe respiratory distress, exhaustion, confusion, drowsiness, or inability to speak full sentences.
- Cyanosis, silent chest, oxygen saturation concern, or rapidly worsening shortness of breath.
- History of intensive care admission, mechanical ventilation, or near-fatal asthma.
- Poor response to reliever medication or increasing need for reliever medication.
- Chest pain, syncope, hemoptysis, suspected foreign body aspiration, or suspected anaphylaxis.

4. Suggested Question Set

- What is the main breathing-related symptom today?
- When did the current episode start, and has it improved, worsened, or fluctuated?
- How often do symptoms occur during the day and at night?
- Are symptoms triggered by exercise, allergens, viral infections, weather changes, smoke, occupational exposures, or pets?
- Has the patient used a reliever inhaler recently? If yes, how often and with what effect?

- Has there been any emergency visit, hospitalization, ICU admission, or intubation for asthma or breathing problems?
- Is there a history of allergic rhinitis, eczema, food allergy, or other atopic disease?
- Does anyone smoke or vape in the home, car, school, or workplace environment?
- Which regular medications are used, and is inhaler technique checked?
- Are there any medication allergies or adverse reactions?

5. Structured Output Guidance for HL7 FHIR JSON

When generating machine-readable anamnesis output, the recommended representation is an HL7 FHIR QuestionnaireResponse resource. Each anamnesis domain should be represented as an item with a stable linkId, human-readable text, and one or more answer values. Patient identity should not be generated by the language model. The application layer should provide the subject reference, encounter reference, timestamps, and provenance metadata.

Recommended QuestionnaireResponse Items

linkId	text
chief-complaint	Chief complaint
history-present-illness	History of present illness
symptom-onset	Symptom onset
symptom-duration	Symptom duration
symptom-frequency	Symptom frequency
triggers	Potential triggers
severity-red-flags	Severity indicators and red flags
past-medical-history	Past medical history
medication-history	Medication and inhaler history
allergy-history	Allergy history
family-history	Family history
social-environmental-history	Social and environmental history
review-of-systems	Review of systems
clinical-summary	Clinical summary for physician review

6. Example Clinical Summary Template

The patient reports symptoms compatible with an asthma-focused clinical history, including [main symptoms]. Symptoms began [onset] and have persisted for [duration]. Reported triggers include [triggers]. Severity markers include [red flags or absence of red flags]. Past history is notable for [asthma/atopy/hospitalization history]. Current medication use includes [medications], with reported adherence and inhaler technique status as [details]. Environmental risk factors include [smoke, allergens, occupational or home exposures]. This AI-generated anamnesis draft requires clinician review and validation.

7. Safety and Governance Notes

Generated anamnesis content should be labeled as AI-assisted draft documentation. The system should preserve source references retrieved from the PDF, apply schema validation, store audit logs, and route outputs to a licensed clinician for review. The model should not autonomously diagnose, prescribe, or decide triage level.

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